

Shiv Shankar Singh

+91-7068900427 | singhshiv0427@gmail.com | Portfolio | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Full-stack developer (React, Node.js, TypeScript) building AI backends from scratch. Built and deployed DocMind, a RAG app with pgvector search, SSE streaming and Zod-validated LLM output, written without LangChain or an LLM SDK to understand how each piece works. Self-taught since graduating in 2024, with a 400+ day GitHub streak and 100+ days of structured DSA practice.

PROJECTS

DocMind: RAG Document Chat | Live App | API | GitHub 2026

TypeScript, Express 5, PostgreSQL, pgvector, Drizzle ORM, Gemini API, Zod, SSE, React 19

- Built the RAG pipeline from scratch, PDFs are chunked, embedded with Gemini and stored in Postgres with pgvector, and each question is answered from the closest chunks with sources shown under every reply.
- Streamed answers word by word over Server-Sent Events by parsing Gemini's raw byte stream, with conversation memory in Postgres so follow-ups and page refreshes keep the same chat.
- Generated quizzes with Gemini structured output validated by Zod (retry once, then fail clean), and shuffled options in code after measuring the model put the right answer in one slot 36% of the time.
- Layered as routes, services and repositories so only repositories touch SQL and services never touch req/res.

TaskForge: REST API & Task Manager | Live App | API Docs | GitHub 2026

Node.js, Express 5, MongoDB, Mongoose, JWT, Zod, Swagger, GitHub Actions

- Built a secure REST API (Express 5, MVC) with JWT auth, bcrypt hashing and Zod validation on bodies and query params, with every MongoDB query scoped to the user from the token so no one sees someone else's tasks.
- Task CRUD with server-side filtering, sorting, pagination, Swagger docs and a 12-check API test script.
- Hardened with Helmet, proxy-aware IP rate limiting (100 req/15 min) and CORS locked to the frontend, with GitHub Actions CI and a Render blueprint deploy with health checks.

Cinegraph: AI Recommendation Engine | Live App | GitHub 2026

Gemini API, Cloudflare Workers, Firebase, TMDB API, React 19, Redux Toolkit

- Built a serverless LLM proxy on Cloudflare Workers so the Gemini key never reaches the browser, with a CORS allow-list and input caps (300-char profile, max 5 history turns) to bound cost and abuse.
- Prompted Gemini for strict JSON and checked every title against TMDB, so made-up titles never reach the UI.
- Personalized picks with a taste profile from user ratings (genre scores, avoided genres, favorite decade) in the prompt, and follow-ups like "more like the third one" run as real multi-turn chat.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), SQL

Backend: Node.js, Express.js, REST APIs, SSE, JWT, bcrypt, Zod, Swagger/OpenAPI, Helmet, Rate Limiting

AI / LLM: Gemini API (generation, streaming, embeddings), RAG, Vector Search, Structured Output

Databases: PostgreSQL, pgvector, Drizzle ORM, MongoDB, Mongoose, Firebase Firestore

Frontend: React.js (v19), Redux Toolkit, Tailwind CSS, Vite

DevOps & Tools: Git, GitHub Actions, Render, Vercel, Netlify, Cloudflare Workers, Neon

EDUCATION

Lovely Professional University

Punjab, India

B.Tech in Computer Science & Engineering, CGPA: 7.01

Oct 2020 – Jul 2024

Tulsi Vidya Niketan

Varanasi, India

Higher Secondary (CBSE Board), 83%

Mar 2018 – Jul 2020

CERTIFICATIONS

Web Development Bootcamp, Udemy (Jul 2023) | **Server-side JavaScript with Node.js**, Coursera (May 2023)

DSA Self Paced, GeeksforGeeks (Jul 2022) | **Java Programming**, Coding Ninjas (May 2022)